

STA623 - Bayesian Data Analysis

Practical 2: Priors, conjugate models, posterior summaries & prediction

Marc Henrion

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Goals. Use conjugate results; check priors by prior predictive simulation; summarise posteriors with point estimates, credible intervals and probabilities; obtain posterior predictive distributions by simulation.

Exercise 1: comparing two groups (General Social Survey)

In the 1990s the General Social Survey recorded the number of children of women aged 40. Women without a college degree: $n_1 = 111$, $\sum_i y_{i,1} = 217$. Women with a degree: $n_2 = 44$, $\sum_i y_{i,2} = 66$. Assume $Y_{i,j} \mid \theta_j \sim_{\text{iid}} \text{Pois}(\theta_j)$ and $\theta_1, \theta_2 \sim_{\text{iid}} \text{Gamma}(2, 1)$.

- (a) Write down the two posterior distributions. Plot them on one graph together with the prior.
- (b) Compute posterior means, medians and 95% equal-tailed and HPD intervals for θ_1 and θ_2 .
- (c) Estimate $P(\theta_1 > \theta_2 \mid \mathbf{y})$ and the posterior median and 95% interval of the rate ratio θ_1/θ_2 by simulation.
- (d) Using simulation, obtain the posterior predictive distribution of the number of children of a new woman in each group. Check group 1 against the exact negative binomial result. Estimate $P(\tilde{Y}_1 > \tilde{Y}_2 \mid \mathbf{y})$ and compare with (c). Why is it so much smaller?

Exercise 2: prior predictive checks

- (a) For the $\text{Gamma}(2, 1)$ prior of Exercise 1, simulate 10^4 values of θ and, for each, the number of children of one woman. What proportion of simulated women have more than 10 children? Does the prior seem reasonable?
- (b) Repeat with the “vague” $\text{Gamma}(0.01, 0.01)$ prior. Comment.
- (c) Suggest a prior for θ that encodes “on average between 1 and 5 children, most likely around 2-3”, and check it by prior predictive simulation.

Exercise 3: posterior summaries for a proportion

$\pi \sim \text{Beta}(2, 3)$ and $Y_1, \dots, Y_{25} \mid \pi \sim_{\text{iid}} \text{Bernoulli}(\pi)$ with $k = \sum y_i = 16$.

- (a) Find the posterior; plot it against the prior.
- (b) Compute the posterior mean, median and mode. Which loss function justifies each?
- (c) Compute 95% equal-tailed and HPD intervals.
- (d) Compute $P(\pi > 0.5 \mid k)$.
- (e) For $H_1 : \pi \in [0.3, 0.5]$ and $H_2 : \pi \in [0.5, 0.7]$ compute the prior odds, posterior odds and Bayes factor BF_{21} .
- (f) Compute the posterior predictive probability that the next observation is a success, analytically and by simulation.

Exercise 4: prior sensitivity and elicitation

For the data of Exercise 3, a clinician tells you that from experience “the proportion is probably around 40%, and I’d be as confident about that as if I’d seen 20 patients”.

- (a) Turn this into a Beta prior.
- (b) Compare posterior mean, 95% equal-tailed interval and $P(\pi > 0.5 \mid k)$ under: $\text{Beta}(1, 1)$, $\text{Beta}(\frac{1}{2}, \frac{1}{2})$, $\text{Beta}(2, 3)$, the clinician’s prior, and a very strong version of it (worth 200 patients). Summarise in a table and a plot.
- (c) Write two sentences you could put in a report about the sensitivity of the conclusions.

Exercise 5: priors on transformed scales

- (a) Simulate 10^5 draws of $\pi \sim \text{Uniform}(0, 1)$ and plot the implied distribution of $\text{logit}(\pi)$ and of the odds $\pi/(1 - \pi)$.
- (b) In a logistic regression, a log odds ratio β is often given the prior $\mathcal{N}(0, 10^2)$. What does this imply for the odds ratio e^β ? Compute $P(e^\beta > 100) + P(e^\beta < 0.01)$. Compare with $\mathcal{N}(0, 1)$.

Exercise 6

- (a) Derive the Jeffreys prior for a Poisson sampling model with n observations. Is it proper? Is the posterior proper?
- (b) Find the conjugate prior for (i) a $\text{Geometric}(\pi)$ sampling model, $p(y \mid \pi) = (1 - \pi)^y \pi$, $y = 0, 1, \dots$; (ii) an $\text{Exponential}(\lambda)$ sampling model.
- (c) Show that the posterior predictive distribution for the Gamma-Poisson model is negative binomial.